

Case Study: Electrical Testing & Reporting Automation

Important: Performance and ROI figures below are illustrative projections based on reasonable workflow assumptions. They must be validated against actual operational data before publication.

Executive Summary

We developed a desktop application for Sarox Technology Inc. that brings electrical test recording, diagnostic calculations, report review, and PDF/Excel export into one controlled workflow.

The application is designed to reduce repetitive report preparation, improve consistency, and help technicians spend more time testing equipment instead of formatting documents.

Projected outcome	Estimated result
Reduction in manual reporting work	65%
Time saved per technician	18–28 hours per month
Faster report completion	55%
Reduction in reporting errors	40%

The Approach

Project Overview

Project detail	Information
Client	Sarox Technology Inc.

Industry	Electrical testing and equipment diagnostics
Project	Winding and Insulation Tester Desktop Application
Duration	7 weeks
Platform	Windows, macOS, and Linux desktop application
Primary users	Electrical testing technicians and report reviewers
Scope	Test recording, calculations, local storage, report overview, instrument communication, and PDF/Excel export

Sarox Technology Inc. performs electrical testing and prepares technical reports that help customers understand the condition of motors and other electrical equipment.

The project involved developing a desktop application that could manage the complete testing workflow—from creating an equipment record and collecting readings to reviewing calculated results and generating a professional report.

The application covers insulation testing, winding measurements, diagnostic calculations, charts, condition ratings, historical record storage, and report export.

The Challenge

Electrical testing generates a large amount of technical information. For every job, technicians may need to:

- Enter customer and equipment information.
- Record readings from insulation and LCR testing instruments.
- Organize multiple test runs.
- Calculate diagnostic ratios and corrected values.
- Compare measurements across electrical phases.
- Identify abnormal imbalance.
- Build charts and summary tables.
- Assign condition ratings.
- Format the final customer report.

When these activities are distributed across instrument displays, handwritten notes, spreadsheets, and document templates, the same information may be handled several times.

That creates four business problems.

Repetitive work

Technicians spend time transferring readings and equipment information between different tools instead of performing higher-value testing and analysis.

Reporting delays

The report cannot be completed until measurements, calculations, charts, and document formatting have all been prepared.

Inconsistent output

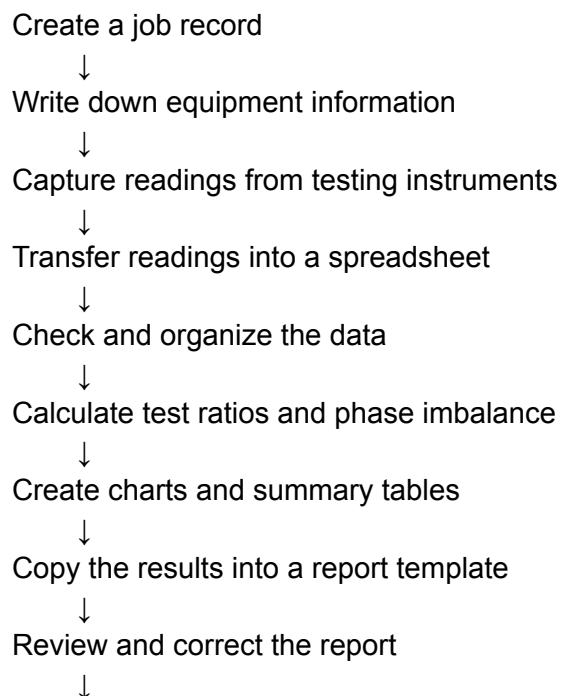
Calculations, terminology, formatting, and condition assessments may vary depending on who prepares the report.

Limited capacity

As testing volume increases, report preparation increases at almost the same rate. Handling more work may therefore require more technician or administrative time.

Before Automation

A typical manual reporting workflow would involve:



Save and deliver the final document

Under the illustrative baseline used for this case study, each report required approximately **80 minutes** of preparation after testing.

At 30 reports per month, this represented approximately **50 technician hours** spent on report preparation. It also introduced multiple opportunities for transcription errors, inconsistent calculations, and formatting corrections.

The Objective

The application needed to achieve six practical objectives:

1. **Create one reliable record** for customer, project, equipment, and test information.
2. **Reduce repeated data entry** by reusing stored information throughout the workflow.
3. **Apply calculations consistently** across the application, PDF reports, and Excel workbooks.
4. **Make technical results easier to review** through summaries, charts, condition ratings, and polar plots.
5. **Generate professional reports quickly** without rebuilding them manually.
6. **Preserve technician control** over testing, interpretation, review, and final report approval.

The application also needed to work locally because electrical testing may take place in environments without dependable internet access.

Our Approach

We organized the application around the technician's real working sequence.

First, we separated the workflow into three categories.

Steps suitable for automation

- Reusing project and equipment information
- Storing readings
- Formatting electrical values
- Calculating ratios and imbalance
- Applying approved rating rules
- Creating charts

- Building report tables
- Generating PDF and Excel files

Steps requiring technician control

- Connecting and configuring test instruments
- Confirming equipment details
- Selecting the correct test
- Reviewing unusual readings
- Interpreting exceptions
- Approving the final report

Steps requiring operational safeguards

- Detecting incomplete or invalid information
- Handling overloaded measurements
- Preventing inconsistent report values
- Preserving records between sessions
- Managing instrument connection failures
- Allowing reports to be reviewed before export

This approach automated repetitive processing without removing professional judgment from the technician.

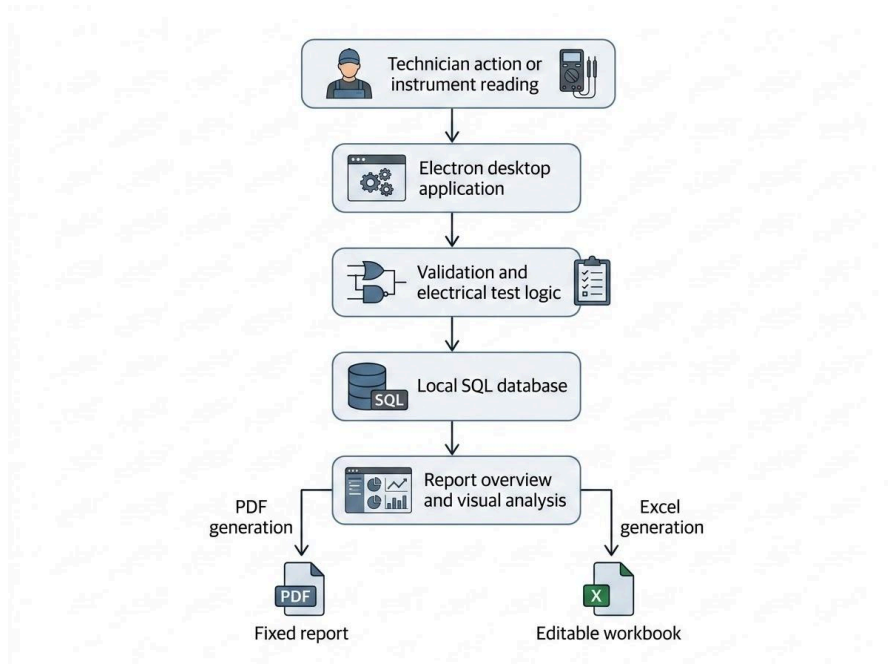
The Solution

We developed a desktop application that combines electrical test recording, data management, analysis, visualization, and reporting.

Technicians can create an equipment record, enter project information, capture insulation and winding measurements, review calculated results, and export the completed report as PDF or Excel.

The application stores information locally and uses the same record to generate the report overview and both export formats. This avoids rebuilding the report from separate files.

Technical Architecture



The application does not require a cloud platform or external CRM to complete its core workflow.

How the Automation Works

Step 1: A test record is created

Trigger: The technician selects “New Record.”

What happens: The application creates a unique record for the job.

System involved: React desktop interface and local SQL database.

What happens next: Customer, project, equipment, and testing information can be added to the record.

Step 2: Equipment information is captured

Trigger: The technician enters project and equipment details.

What happens: Information such as the client, equipment configuration, testing location, insulation class, wire markings, and test voltages is saved.

System involved: Information form and local database.

What happens next: The information becomes available to the test screens and report generator without being entered again.

Step 3: Insulation readings are collected

Trigger: The technician starts an insulation test or receives readings from the supported instrument workflow.

What happens: Time, applied voltage, actual voltage, current, and resistance values are recorded against the selected test run.

System involved: Insulation-testing interface, serial communication layer, and database.

What happens next: The data becomes available for calculation, graphing, and reporting.

Step 4: Winding measurements are recorded

Trigger: The technician selects a multimeter field or sends a supported measurement command.

What happens: The application stores the measured value together with relevant temperature and frequency information.

System involved: Multimeter interface, serial communication layer, and database.

What happens next: Phase measurements can be compared and used in report summaries.

Step 5: Diagnostic values are calculated

Trigger: Sufficient test data becomes available.

What happens: The application calculates relevant results, including Polarization Index, Dielectric Absorption Ratio, corrected insulation resistance, and phase imbalance.

It also assigns consistent condition bands:

- Excellent
- Good
- Normal
- Observe
- Caution
- Alarm

System involved: Application calculation and rating logic.

What happens next: Results appear in the report overview with the related tables and charts.

Step 6: The technician reviews the report

Trigger: The technician opens the Report screen.

What happens: The application retrieves the equipment information, insulation results, winding measurements, calculations, ratings, and visualizations.

System involved: Report overview and local database.

What happens next: The technician can check the report before creating the final file.

Step 7: The report is exported

Trigger: The technician selects “Export PDF” or “Export Excel.”

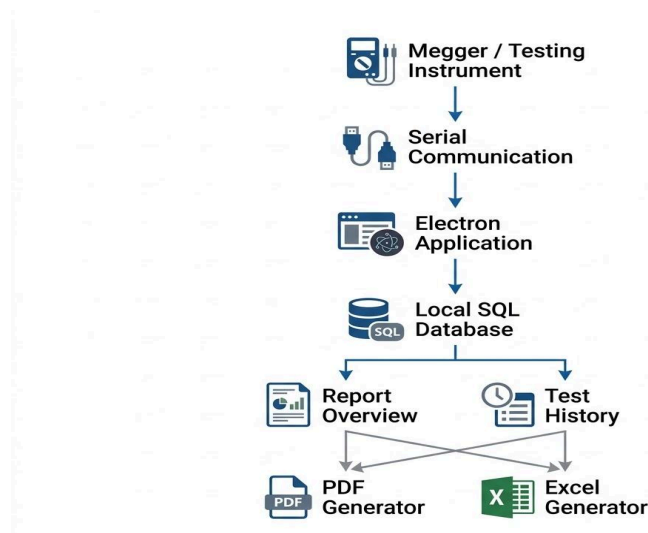
What happens: The application formats the available data, calculations, charts, ratings, and branding into the selected report format.

System involved: PDFKit for PDF generation and ExcelJS for Excel generation.

What happens next: A save dialog allows the technician to choose the final file location.

Integrations

The current application uses local software components and direct instrument communication rather than external cloud services.



Integration	Purpose	Communication method
Megger testing instrument	Collect insulation-test readings	Serial connection
LCR multimeter	Collect winding-related measurements	Serial connection and device commands
Local database	Store records and test measurements	Internal SQL operations
PDFKit	Generate fixed-format reports	Internal application library
ExcelJS	Generate editable workbooks	Internal application library
Operating system	Select report locations and open files	Electron desktop APIs

Instrument configuration and data parsing must match the exact protocol of the physical devices used in production.

Technical Challenges

Challenge 1: Maintaining consistent results across three outputs

The same measurement appears on the report screen, in the PDF report, and in the Excel workbook. If each area calculates results differently, customers could receive conflicting information.

Solution: Calculation, formatting, and rating rules were standardized across the reporting workflow. This includes resistance formatting, phase imbalance, temperature correction, test ratios, and condition bands.

Challenge 2: Presenting complex electrical data clearly

Raw electrical readings are necessary for technical review, but they are difficult for many customers and business stakeholders to interpret quickly.

Solution: The application presents the data through summary tables, trend charts, phase comparisons, polar plots, calculated ratios, and color-coded condition ratings. Readers can see both the underlying readings and the overall assessment.

Challenge 3: Generating reports with variable amounts of data

Different jobs may contain different test types, numbers of readings, charts, logos, and equipment details. A rigid report template may produce empty sections or broken layouts.

Solution: Report sections are created dynamically from the information available in each record. The PDF and Excel exporters include the relevant tests and visualizations without requiring the technician to rebuild the document.

Challenge 4: Supporting offline work

Testing locations may not always have reliable internet access. A cloud-dependent application could interrupt field operations.

Solution: Equipment records and measurements are stored locally. Core testing, review, and report generation can continue without an internet connection.

Challenge 5: Handling physical instrument communication

Serial devices can return partial messages, unexpected formats, or connection errors.

Solution: Instrument communication was separated into a dedicated serial layer. This layer manages port discovery, connection status, incoming data, parsing, commands, disconnection, and simulator support.

Before vs. After

Area	Before	After
Manual work	Readings, calculations, charts, and report content prepared separately	Data reused automatically throughout the application
Processing time	Approximately 100 minutes per report	Approximately 45 minutes per report
Errors	About 10 transcription or calculation corrections per month	About 6 corrections per month

Data entry	Information entered into multiple files or templates	Information entered once per equipment record
Calculations	Prepared or checked separately	Calculated consistently by the application
Charts	Created manually or in a separate spreadsheet	Generated from stored measurements
Notifications/actions	Report manually assembled and located	PDF or Excel generated directly from the reviewed record
Scalability	About 22 completed reports per technician each month	About 30 completed reports per technician each month
Record storage	Dependent on individual file organization	Structured local record storage
Internet dependency	Dependent on tools used previously	Core application operates offline

Projected Results

Repeated data transfer, calculations, chart preparation, and document formatting are reduced through one connected workflow.

At approximately 30 reports per month, the projected time saving is between 18 and 28 hours, depending on report complexity.

Projected report-preparation time decreases from approximately 100 minutes to 45 minutes.

Consistent calculations and reduced transcription are projected to lower calculation, copying, and formatting corrections.

Projected capacity increases from approximately 22 to 30 reports per technician each month without a proportional increase in administrative work.

Conclusion

More productive technician time

Reducing repetitive report preparation gives technicians more time for testing, diagnosis, customer communication, and exception review.

Faster customer response

Reports can be generated as soon as the test record has been reviewed. Customers receive the information needed for maintenance decisions sooner.

Lower administrative cost

Automation reduces the paid technical time spent copying readings, checking formulas, recreating charts, and formatting reports.

More consistent operations

Approved calculation and rating rules are applied through the same application workflow, helping the company maintain a repeatable reporting standard.

Increased capacity

Reducing preparation time allows the same team to complete more reports without increasing administrative effort at the same rate.

Reduced dependency on individual spreadsheets

The reporting process is stored within the application rather than depending on one employee's spreadsheet formulas, templates, or document-formatting knowledge.

Better customer communication

Charts, summaries, and condition ratings help decision-makers understand test results without removing the technical detail required by engineers.

Technology Stack

Component	Technology	Purpose
Desktop application	Electron	Runs the application as installable desktop software
User interface	React	Provides the testing and reporting screens
Navigation	React Router	Manages application views
Local database	SQL.js	Stores equipment records and test measurements
Instrument communication	SerialPort	Connects with supported electrical testing instruments
Charts	Recharts	Displays test trends and comparisons
Custom visualization	SVG	Produces impedance polar plots
PDF reporting	PDFKit	Generates customer-ready PDF reports
Spreadsheet reporting	ExcelJS	Generates structured Excel workbooks
Application packaging	Electron Forge	Builds distributable application packages
Build system	Webpack and Babel	Compiles the desktop application

Key Takeaways

- Effective automation begins with the operating process, not the report template.
- Electrical readings, calculations, review, and export can be managed within one controlled application.
- Reusing stored information reduces duplicate entry and reporting effort.
- Technician judgment remains part of testing, exception handling, and final approval.
- Local storage allows core work to continue without dependable internet access.
- Automated PDF and Excel generation can improve speed without sacrificing technical detail.